Thermally Reversible and Microscopically Tunable Gelation in Recombinant Polypeptides

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Abstract

Elastin-like polypeptides gel on heating, and whether the gel dissolves again on cooling decides whether it can serve as a switchable material. We followed the thermally induced gelation of concentrated aqueous $(\mathrm{VPAVG})_x$, with $x = 10$, 20 and 30, by small-angle X-ray photon correlation spectroscopy (SA-XPCS) during heating at $0.1\,^{\circ}\mathrm{C\,min^{-1}}$. Each 2 s exposure was taken at a fresh sample position, and heating stopped once the intensity autocorrelation function no longer decayed. All three develop a two-step decay before arresting, with a fast mode that carries 27–34% of the amplitude. For $(\mathrm{VPAVG})_{30}$

the SAXS profiles collapse onto one master curve up to 29.6 °C, and the characteristic length grows as $(T_0 - T)^{-0.80}$ toward $T_0 = 30.5$ °C. Reversibility depends on chain length. $(VPAVG)_{10}$ and $(VPAVG)_{20}$ stay arrested on cooling, whereas $(VPAVG)_{30}$ largely recovers during cooling and returns to its dispersed state after storage at 4 °C.

Introduction

Genetically engineered biopolymers such as elastin-like polypeptides (ELPs) are a versatile platform for stimuli-responsive soft materials.[1,2] These repetitive protein polymers, derived from the hydrophobic motifs of tropoelastin, show a tunable lower critical solution temperature (LCST) transition.[3] Above a characteristic temperature, ELPs phase separate through hydrophobic association and desolvation.[4,5] The equilibrium thermodynamics of this transition is well documented. The kinetic pathway by which the protein-rich phase arrests into a viscoelastic gel is much less understood. Because the properties of ELP materials are sensitive to sequence, knowing how molecular parameters govern the onset and the reversibility of gelation is central to the design of dynamic biopolymer networks.[6]

The reversibility of an ELP gel is tied to its thermal history. Prolonged exposure to elevated temperature coarsens and densifies the protein-rich network and can carry it into a structurally arrested state that no longer dissolves on cooling.[5] A measurement of the onset of gelation therefore has to be fast compared with the coarsening that follows it, and bulk methods that need long thermal equilibration, such as temperature-dependent rheology, sample the network after it has evolved.

Here we follow the onset with in situ small-angle X-ray photon correlation spectroscopy (SA-XPCS).[7,8] Metallic sample cells of high thermal conductivity and the coherent flux of the Advanced Photon Source allow the sample to be cooled as soon as dynamic arrest is detected. Each measurement is a 2 s exposure, and the beam moves to a fresh spot for the next one. This keeps the dose per spot low while using the full coherent flux, and a detector running at 52 kHz resolves the dynamics down to 20 µs as the network forms.[9,10]

The short window limits the observation to relaxations faster than a few seconds. In arresting colloidal networks studied by XPCS together with rheology, however, the evolution from a two-step decay to a flat intensity autocorrelation function g_2 marks a state past the macroscopic gel point, where $G' > G''$.[11] A flat g_2 is therefore an in situ indicator of a formed gel network.

In this Letter we compare the thermally induced gelation of aqueous $(\mathrm{VPAVG})_x$ with $x = 10$, 20 and 30. The SAXS profiles show the growth of a hierarchical network of the kind formed by viscoelastic phase separation, and the correlation functions show fast local and slow collective modes emerging together before arrest.[12,13] Measured on heating and on cooling under one protocol, the three chain lengths share their gelation dynamics but differ in whether the gel dissolves again.

Results and Discussion

Each sample contained about $100\,\mathrm{mg\,mL^{-1}}$ of recombinant protein in 20 mM HEPES (pH 8), 250 mM NaCl and 2 mM DTT; synthesis and purification are described in the Supporting Information. The solutions were pipetted into aluminum liquid cells (Figure S1), equilibrated at 6 °C, heated to 20 °C at $1\,^{\circ}\mathrm{C\,min^{-1}}$, and then heated at $0.10\,^{\circ}\mathrm{C\,min^{-1}}$ while SA-XPCS was recorded continuously (Figures S2a–S4a). Heating stopped once g_2 no longer decayed (Figures S2e–S4e), at 45.2, 41.4 and 32.8 °C for $(\mathrm{VPAVG})_{10}$, $(\mathrm{VPAVG})_{20}$ and $(\mathrm{VPAVG})_{30}$, respectively. Each sample was then cooled to 20 °C at the same rate (Figures S2b–S4b). The heating ramps comprise 2780, 2389 and 1198 acquisitions, and each cooling ramp the same number to within one. Every acquisition recorded 100,000 frames at 52 kHz at a fresh sample position, so that no spot was exposed twice. The radiation-damage controls are described in the Supporting Information, and the measurement is sketched in Figure 1.

A single 2 s acquisition limits the sensitivity to very slow arrest, which in related systems can develop over much longer times.[14–16] The short exposure, however, allows the high incident

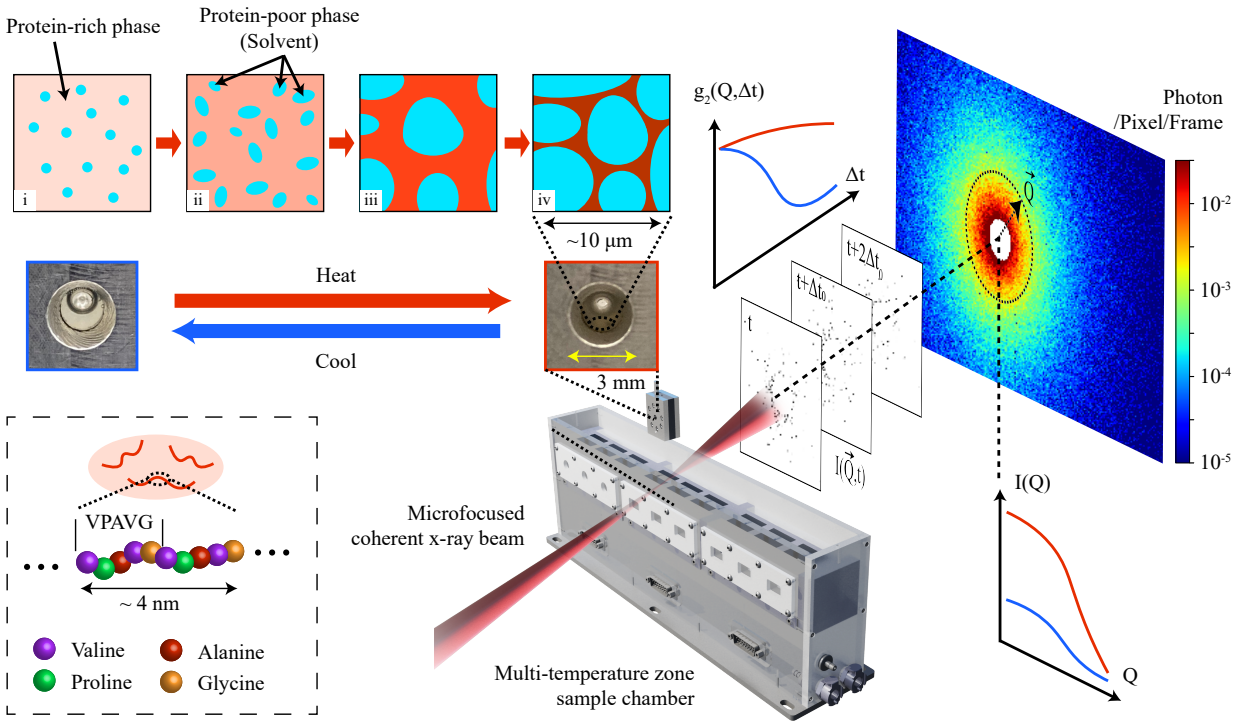


Figure 1: Schematic of the SA-XPCS measurement of temperature-induced gelation in elastin-like polypeptides. A partially coherent X-ray beam of 10.94 keV is focused to 10 µm × 10 µm on the sample, and the coherent scattering pattern $I(\vec{Q}, t)$ ("speckle") is recorded by a photon-counting detector 7.8 m downstream at a frame rate of 52 kHz. $(\text{VPAVG})_x$ solutions (inset, bottom left) are loaded into 2 mm thick aluminum cells with 3 mm diameter polycarbonate windows and held at 6 °C in a multi-zone temperature-controlled chamber. As the temperature rises slowly, the ELPs assemble into a hierarchical, percolating network (top left, stages i–iv). The larger structures raise the SAXS intensity $I(Q)$ at low Q, and the arrest of their dynamics flattens the intensity autocorrelation function $g_2(Q, \Delta t)$, as sketched by the red and blue curves on the right.

flux that is essential for resolving fast dynamics with a photon-counting detector. Even so, the photon count per 19.2 µs frame remains sparse, which makes two-time correlation analysis impractical,[17,18] and g_2 was computed with the multi-tau algorithm. To improve statistics, each ramp was divided into ten consecutive sections of equal acquisition count (twenty for $(\mathrm{VPAVG})_{20}$), and $I(Q)$ and $g_2(Q, \Delta t)$ were averaged within each section. Before averaging, every acquisition was compared with every other in its section by the cross-correlation of its g_2, of the uncertainty of g_2, and of $\log I(Q)$. The least similar 5% on any of the three were excluded, about 12% of the acquisitions in all. The code for the reduction and for every figure is publicly available (Data and Code Availability), as is the original analysis.[19]

Figure 2 compiles the temperature-dependent SAXS profiles $I(Q, T)$ of $(\mathrm{VPAVG})_{30}$ during heating, on the absolute scale with the buffer subtracted (Supporting Information). Between 20.6 and 32.2 °C the intensity at $Q < 0.006\,\text{Å}^{-1}$ rises 190-fold, which indicates a growing population of structures larger than about 100 nm. At the lowest temperatures $I(Q)$ is flat at small Q, as expected for dispersed chains, and a Guinier analysis of the same construct at 12-ID-B gives an apparent radius of gyration $R_g = 22.65 \pm 0.50$ Å (Figure S5b).

On heating, the profiles first show an apparent self-similarity. After rescaling both I and Q by temperature-dependent factors, each profile can be laid onto its predecessor, and the profiles up to 29.6 °C collapse onto a common master curve (inset, Figure 2a). The collapse fails at 30.9 and 32.2 °C. Within the self-similar regime, the Q-rescaling factors multiplied by the length of the dispersed state, R_g, give a characteristic length $\xi(T)$, which grows from 2.3 nm at 20.6 °C to 14 nm at 29.6 °C (Figure 2b). A power law $\xi \propto (T_0 - T)^{\gamma}$ describes this growth with $T_0 = 30.5$ °C and $\gamma = -0.80 \pm 0.04$, so the length diverges at a temperature between the last collapsing profile and the first one that does not.

To rationalize the evolution of $I(Q, T)$, we describe the profiles with a generalized Lorentzian, $I = I_0/[1 + (\xi Q)^{\eta}]$. This form has been used for liquid–liquid phase separation in aqueous–oil–surfactant ternaries approaching criticality,[18] and it is motivated by the model H dynamics of binary liquids.[20] At low and intermediate temperatures it describes the data well,

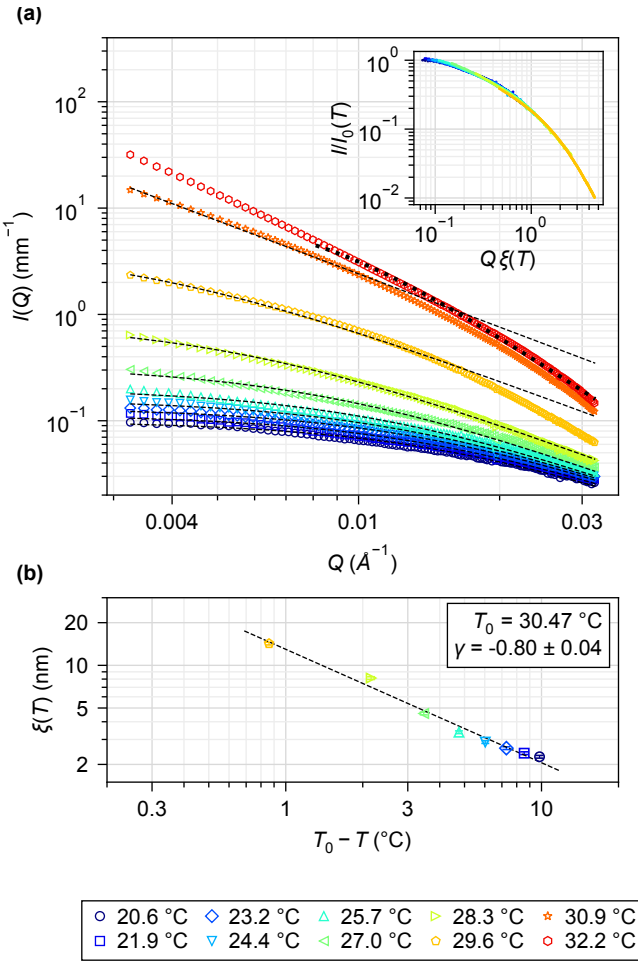


Figure 2: Evolution of structure during thermally induced gelation of $(\mathrm{VPAVG})_{30}$. (a) Buffer-subtracted SAXS intensity $I(Q,T)$ on the absolute scale at the ten temperatures of the heating ramp, each the average of one section of about 120 acquisitions. Dashed lines are fits of the generalized Lorentzian $I = I_0/[1 + (\xi Q)^\eta]$ with $\eta = 1.67$ held fixed, to every profile except the hottest. The dotted line is the same function with η free, fitted to the 32.2 °C profile at $Q \geq 0.008\,\text{Å}^{-1}$, which gives $\eta = 2.92 \pm 0.02$. Inset: the profiles up to 29.6 °C, each rescaled in Q and I onto its predecessor by minimizing the mean squared difference in $\log I$, plotted against $Q\xi(T)$ and normalized by the lowest-Q intensity of the 20.6 °C profile. The fixed η is the slope of $\ln(I_{\max}/I - 1)$ against $\ln Q$ over this master curve, with $I_{\max}$ its highest intensity. (b) Characteristic length $\xi(T) = R_g\, s(T)$, where $s(T)$ is the cumulative Q-rescaling factor and $R_g = 22.65 \pm 0.50$ Å is the Guinier radius of the dispersed chains (Figure S5b), against $T_0 - T$. The dashed line is the least-squares power law $\xi \propto (T_0 - T)^\gamma$ in log–log coordinates, with T_0 chosen to minimize the residual. Error bars are the uncertainty of R_g, which is common to all points. The key identifies the temperatures in both panels.

consistent with the collapse in the inset of Figure 2a, and a linear regression of the master curve gives $\eta = 1.67$. At higher Q and higher temperature, systematic deviations appear and two regimes coexist: a low-Q region with $\eta \approx 1.67$, and a steeper high-Q decay, for which a fit of the 32.2 °C profile gives $\eta = 2.92 \pm 0.02$. Exponents between 1 and 3 are characteristic of mass-fractal aggregates, and the steeper decay indicates a denser structure on the shorter length scales.[21] The departure from a single η, together with the breakdown of the collapse, shows that the network is no longer self-similar across scales and develops a hierarchical architecture. Such a network can support motion on two scales, rapid rearrangement of dense local strands and slow collective motion of the network as a whole, which anticipates the two dynamical modes described next.

The XPCS measurements resolve these modes. Figure 3 shows the evolution of $g_2(Q, \Delta t)$ of $(\mathrm{VPAVG})_{10}$ at $Q = 0.0038\,\text{Å}^{-1}$, a length scale $2\pi/Q$ of 167 nm, during the heating ramp. At 21.3 °C, g_2 decays completely within a few milliseconds, indicating mobile structures. With increasing temperature the dynamics slow down markedly, and at 44.0 °C g_2 falls by less than 0.04 over the whole 2 s window: the network has arrested. At intermediate temperatures, before complete arrest, g_2 decays in two distinct steps.

The two-step decay appears in all three samples and points to coexisting fast and slow dynamics at intermediate stages of gelation. As Figure 4 shows, it is described quantitatively by a superposition of two stretched exponentials.[22,23] For a given Q,

$$g_2(\Delta t) = 1 + \beta\, g_1^2(\Delta t), \tag{1}$$

$$g_1(\Delta t) = a_1 \exp[-(\Delta t/\tau_1)^{p_1}] + (1 - a_1) \exp[-(\Delta t/\tau_2)^{p_2}]\,, \tag{2}$$

where g_1 is the normalized intermediate scattering function, τ_1 and τ_2 are the characteristic times of the fast and slow modes, p_1 and p_2 describe the breadth of their distributions, and a_1 is the fraction of the decay carried by the fast mode. The contrast β is set by the coherence of the beam and was taken for each sample from the short-delay level of g_2. The

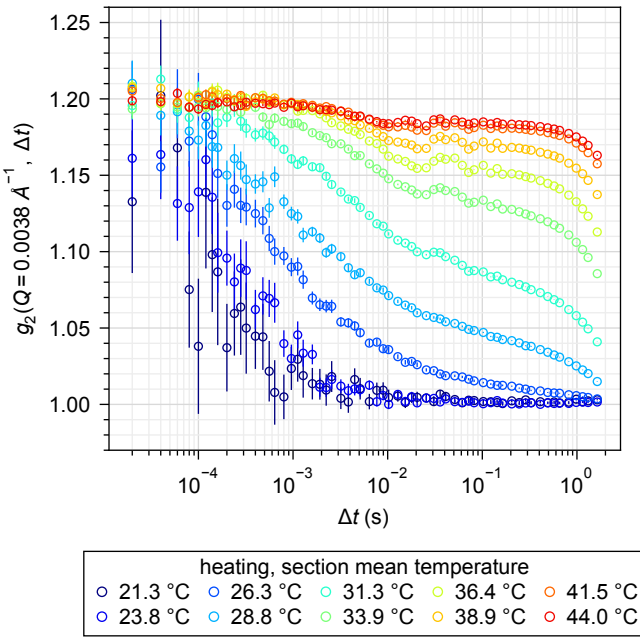
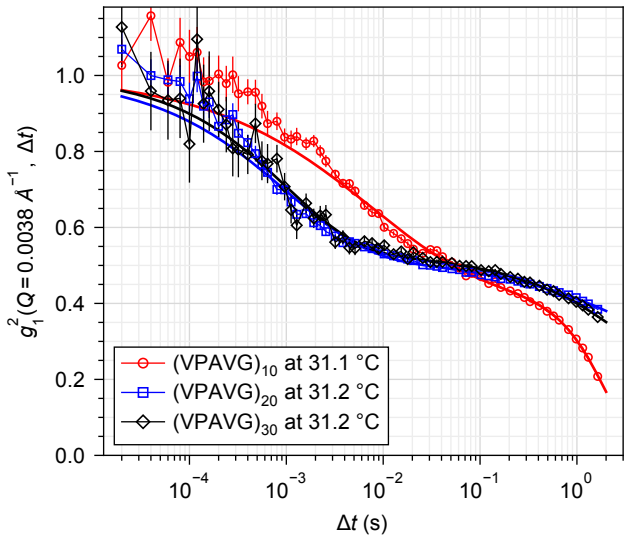


Figure 3: Arrest of dynamics during thermally induced gelation of $(\mathrm{VPAVG})_{10}$. Intensity autocorrelation functions $g_2(Q = 0.0038\,\text{Å}^{-1}, \Delta t)$ during heating at $0.10\,^{\circ}\mathrm{C\,min^{-1}}$ from 20.0 to 45.2 °C. Each curve averages one section of 278 consecutive acquisitions spanning 2.5 °C, and the key gives the mean temperature of each section. At low temperature g_2 decays completely within milliseconds; at the highest temperatures it barely decays within the 2 s window, the signature of an arrested gel.

fitted parameters are listed below Figure 4. The fast mode of $(\mathrm{VPAVG})_{10}$ is an order of magnitude slower than those of $(\mathrm{VPAVG})_{20}$ and $(\mathrm{VPAVG})_{30}$, 11 ms against 1.2–1.3 ms, while its weight a_1 lies in a similar range, 27–34%, in all three. The slow times of $(\mathrm{VPAVG})_{20}$ and $(\mathrm{VPAVG})_{30}$ lie far outside the measurement window and serve only as effective descriptors of the long-lived component. Furthermore, the dynamics of $(\mathrm{VPAVG})_{20}$ and $(\mathrm{VPAVG})_{30}$ at the same temperature nearly coincide, even though $(\mathrm{VPAVG})_{20}$ was eventually heated 8.6 °C further before it arrested (Figures S3 and S4). This indicates a common dynamic process during the gelation of these ELPs.



	a_1	τ_1 (ms)	τ_2 (s)	p_1	p_2
$(\mathrm{VPAVG})_{10}$	0.337 ± 0.006	11.1 ± 0.9	3.4 ± 0.1	0.45 ± 0.02	1.36 ± 0.07
$(\mathrm{VPAVG})_{20}$	0.274 ± 0.004	1.3 ± 0.1	163.9 ± 44.0	0.55 ± 0.03	0.41 ± 0.03
$(\mathrm{VPAVG})_{30}$	0.268 ± 0.003	1.2 ± 0.1	37.4 ± 4.1	0.62 ± 0.06	0.53 ± 0.02

Figure 4: Two-step decay in the dynamics of $(\mathrm{VPAVG})_{10}$, $(\mathrm{VPAVG})_{20}$ and $(\mathrm{VPAVG})_{30}$ near 31.2 °C. The plotted quantity is $g_1^2 = (g_2 - 1)/\beta$ at $Q = 0.0038\,\text{Å}^{-1}$, with $\beta = 0.19$, 0.195 and 0.135 from the short-delay level of g_2 of each sample. Solid lines are fits of $g_1 = a_1 \exp[-(\Delta t/\tau_1)^{p_1}] + (1 - a_1) \exp[-(\Delta t/\tau_2)^{p_2}]$ weighted by the measured uncertainties, and the table lists the fitted parameters. Each curve averages one section of its heating ramp: 120, 107 and 52 acquisitions.

Despite the similar dynamics during gelation, the reversibility of the gel depends strongly on chain length (Figures S2–S4). The structures formed by $(\mathrm{VPAVG})_{10}$ do not dissolve during

the cooling ramp: $I(Q)$ and g_2 remain locked in. At the end of cooling the low-Q intensity is still 17 times its value at the start of heating, and g_2 stays flat. $(\mathrm{VPAVG})_{20}$ behaves the same way, ending at 27 times its initial intensity. The irreversibility of $(\mathrm{VPAVG})_{10}$ persists well beyond the SA-XPCS time window: the cell remained optically opaque after more than 48 h at 4 °C (Figure S7). $(\mathrm{VPAVG})_{30}$, in contrast, recovers much of its structure and dynamics during cooling. Its low-Q intensity falls from 190 to 5 times the initial value, and below about 24 °C g_2 decays again within the window. A separate loading, heated at 1 °C min^{-1} and cooled to 6 °C between heatings, was taken through seven consecutive cycles to 34 °C (Figure S6). It arrests on every heating, and when it is next measured at 6 °C, 7 to 41 min after the end of each ramp, it is back at nearly the same profile: over the seven cycles the low-Q intensity at 6 °C varies by 9%, against a 140-fold rise at the top of every ramp. After storage at 4 °C for more than 24 h the recovery is complete.

At 12-ID-B, an aliquot that had been through more than ten heating and cooling cycles gives the same 10 °C profile as an aliquot that was never heated, the dispersed state of $R_g = 22.65$ Å (Figure S5a). On heating to 30 °C, both aliquots develop two broad wide-angle peaks, at 0.72 and 1.38 Å^{-1}, which are absent again at 10 °C. Their spacings, 8.8 and 4.5 Å, are compatible with the sheet-to-sheet and strand-to-strand distances of cross-β-like packing.[24,25] The peaks are broad, corresponding to correlation lengths of only 19 and 25 Å, so the ordered contacts extend over a few repeat spacings. Thermally labile β-sheet-like contacts of this kind are plausible physical junctions of a protein network,[26–28] and in $(\mathrm{VPAVG})_{30}$ they form on heating and dissolve on cooling together with the rest of the assembly, so the thermal history is erased. Video S1 supports the same conclusion: touching a capillary of $(\mathrm{VPAVG})_{30}$ triggers gelation, and the structures dissolve within about a minute once the heat source is removed.

We interpret the thermally induced gelation of $(\mathrm{VPAVG})_x$ at these high protein concentrations, about 100 mg mL^{-1} or 10% by volume, as arrested phase separation,[5] and more specifically as gelation driven by viscoelastic phase separation.[12] The stages of this process

are sketched in Figure 1. In Tanaka's framework for dynamically asymmetric mixtures, the viscosity of the solvent is much lower than that of the protein phase, so the solvent separates out first as "solvent holes" (stage i). As phase separation progresses, the holes grow and impinge (stage ii), producing a spatially percolating protein-rich skeleton (stage iii). Continued heating through the LCST strengthens hydrophobic association, expels more water and densifies the skeleton until interfacial stresses halt coarsening (stage iv).[13] This pathway is consistent with the SAXS. At first the profiles collapse onto a single-length scaling form without a structure-factor peak in the measured Q window. At higher temperature the collapse breaks down into a low-Q region with $\eta \approx 1.7$ and a steeper high-Q region with $\eta \approx 2.9$, the hierarchical connectivity also seen in other networks of thermoresponsive ELPs formed by arrested phase separation.[29] The microscopic driving force is the increase of side-chain hydrophobicity with temperature[30] and the accompanying desolvation-driven association of the VPAVG segments,[5] which bias the system toward densification and arrest.

The dynamics resolved by XPCS are consistent with this structural evolution. As the gel forms, g_2 develops a pronounced two-step decay: a fast relaxation at short delays followed by a much slower decay that approaches arrest. In associative protein networks, single-chain relaxation proceeds through Zimm-like segmental motion and subsequent center-of-mass diffusion; neutron spin-echo measurements place the Zimm crossover at nanoseconds, with a high-Q relaxation time that scales as Q^{-3}.[9] The fast relaxation seen by XPCS appears at far longer delays, 1–10 ms. Together with the hierarchical structure resolved by SAXS, this points to a fast mode arising from local rearrangement of fibrils or microdomains within the densifying protein skeleton, which decorrelates the speckle without requiring escape of whole chains. We attribute it to this reconfiguration because it slows continuously while the SAXS shows the network developing. The slower mode corresponds to collective, junction-mediated relaxation and the eventual arrest of the network. A similar decoupling of local and collective dynamics has been reported in Laponite clay gels[10] and in the pressure-induced gelation of lysozyme.[31] The fast and slow modes therefore reflect an interplay between local stress

relaxation and large-scale consolidation of the network, which marks the transition from fluid-like to arrested viscoelastic behavior.

In summary, SA-XPCS with short exposures at fresh positions follows the thermally induced gelation of three sequence-defined ELPs through the formation of a hierarchical network and the arrest of its dynamics. The gelation pathway is shared by all three chain lengths, while the reversibility of the gel is set by chain length: ten and twenty VPAVG repeats remain arrested on cooling, and thirty repeats return to the dispersed state.

Methods

Samples. ELPs with 10, 20 and 30 repeats of the pentapeptide VPAVG were cloned by overlap-extension rolling-circle amplification,[1] expressed in *E. coli*, purified by immobilized metal affinity chromatography, and exchanged into 20 mM HEPES (pH 8.0), 250 mM NaCl and 2 mM DTT at about 100 mg mL^{-1}. The procedures are given in the Supporting Information.

SA-XPCS. Measurements were made at beamline 8-ID-I of the Advanced Photon Source, $(VPAVG)_{10}$ and $(VPAVG)_{20}$ in July 2021 and $(VPAVG)_{30}$ in March 2022, with the same cell, stage and protocol. A partially coherent beam of 10.94 keV with an incident flux of $1.6–1.9 \times 10^{10}$ photons/s was focused to about 10 µm × 10 µm at the sample. Scattering was recorded 7.8 m downstream by a photon-counting detector (XSPA-500k)[32] at 52 kHz, 100,000 frames per acquisition, every acquisition at a previously unexposed position. g_2 was computed with the multi-tau algorithm, and $I(Q)$ by temporal and azimuthal averaging; the absolute-scale calibration, the averaging and the radiation-damage controls are described in the Supporting Information.

SAXS/WAXS. Two aliquots of $(VPAVG)_{30}$ in 2 mm quartz capillaries were measured at 10 °C and 30 °C at beamline 12-ID-B of the Advanced Photon Source by simultaneous SAXS and WAXS at 13.3 keV.[33–35]

Supporting Information Available

Recombinant synthesis and purification, and the oligonucleotide sequences; sample cells and the automated cell assembly; SA-XPCS reduction and absolute-intensity calibration; structure and dynamics of all three chain lengths on heating and cooling; 12-ID-B reversibility, wide-angle peak fits and Guinier analysis; seven heating and cooling cycles of $(VPAVG)_{30}$; irreversibility of $(VPAVG)_{10}$; molecular dynamics of a $(VPAVG)_{30}$ array; radiation-damage controls; code and data availability; AI-assisted development (PDF). Videos of the reversible transition in $(VPAVG)_{30}$ and of the robotic cell assembly (MP4).

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AI-assisted manuscript and code preparation. The authors used Anthropic's Claude models, accessed through Argo, Argonne National Laboratory's internal generative AI interface, to assist with manuscript preparation and with development of the data-analysis and figure-generation code. That code and the reduced data it operates on are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP-Reversibility`, so the reported results can be regenerated and checked independently. The authors take full responsibility

for all manuscript content and reported results.

The authors declare no competing financial interest.

Data and Code Availability

Reduced data and analysis code are at `https://github.com/qzhang234/PubCode-2021-IDPP-Reversibility`.

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